

Fine-Tooth Iron-Core Linear Synchronous Motor for Low Acoustic Noise Applications

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Abstract—The design of a novel low-noise high-force linear motor for precision positioning applications, such as in semiconductor lithography, is presented in this paper. New magnetic designs achieve low noise and vibration by reducing high spatial-frequency magnetic field components, which cause harmonic vibrations of a moving stage. To reduce such force harmonic content, our new motor has fine teeth, narrow slots with high slot aspect ratio, five phases, and a moving skewed Halbach magnet array. As compared with a commercially available iron-core motor, in our new design, the acoustic noise level in Pascals is reduced by more than 90%, corresponding to a 20 dB reduction in sound pressure level, even with a nonskewed magnet array. Using a skewed magnet array, the cogging-driven and velocity-dependent noise is further reduced by 90% in Pascals. This new linear motor simultaneously is capable of very high shear force densities in order to support high-acceleration motions, and thus provides new options for high-performance and quiet linear motion devices.

Index Terms—Acoustic noise, iron-core permanent magnet motor, linear motor, multiphase motor.

I. INTRODUCTION

THIS paper presents new designs and experimental results for linear iron-core permanent-magnet motors operating at high velocity and acceleration in machines requiring nanometer-scale precision. Linear iron-core synchronous motors have the potential for high force capability as compared with air-core motors. However, such motors also have the problem of large normal and tangential forces acting between the permanent magnets and the iron core, which thereby introduce vibrations and acoustic noise.

Acoustic noise and vibration of electric motors can be generated by many causes, including magnetic, mechanical, and electronic sources [1], [2]. Among these sources, radial deformation of the stator caused by radial magnetic force is a major source of noise in rotary machines, including variable-reluctance

motors [3], switched reluctance motors [4]–[9], permanent-magnet synchronous motors [10]–[13], and induction motors [14]–[16].

Other sources such as cogging and the associated motor vibration are studied in [17] and [18]. Magnetostriction, causing transformer noise [19], is considered as another noise source in [20] and [21]. However, these results indicate that the contribution of the magnetostriction effect to motor noise is minor. In the mechanical sources of the motor noise, there can be several auxiliary sources other than the vibrational modes of the motor stator [1], [2], [22]. The use of rolling-element bearings requires mechanical contact, thereby possibly causing vibrational and acoustic noise. Rotor unbalance can also be a source of noise in rotary machines. In linear motors, electrical cables attached to the moving stage may cause dynamic interactions resulting in noise and vibration.

There are also potential electronic sources of acoustic noise and vibration. When pulsewidth modulation (PWM) switching drives are used, the high-voltage switching waveform and associated currents may directly generate acoustic noise and vibration [23], [24]. Encoder quantization and interpolation error can also be an electronic noise source for motors under closed-loop control as these signals propagate through the controller into motor forces and torques.

As compared with rotary machines, the noise issue of linear motors has not yet been studied as extensively in the literature. In this paper, the following tasks are performed.

- 1) It identifies and investigates the vibro-acoustic noise of linear iron-core permanent magnet motors.
- 2) It differentiates noise sources between the acceleration and constant-velocity regions.
- 3) It provides a new motor design to reduce such noise for both regions while increasing shear stress performance compared with conventional and commercial linear iron-core motors.
- 4) It experimentally validates the noise source and noise reduction using the proposed magnetic design solution.

The rest of this paper is organized as follows. In Section II, we discuss our hypothesis on the noise causes of linear iron-core motors. In Section III, we study and compare the force harmonic content of a commercially available iron-core motor and our new motor. The magnetic design of our new motor is discussed in Section IV. We then introduce an experimental linear stage testbed and present experimental results on the noise reduction of our new motor compared with a conventional one in Sections V and VI, followed by conclusions in Section VII.

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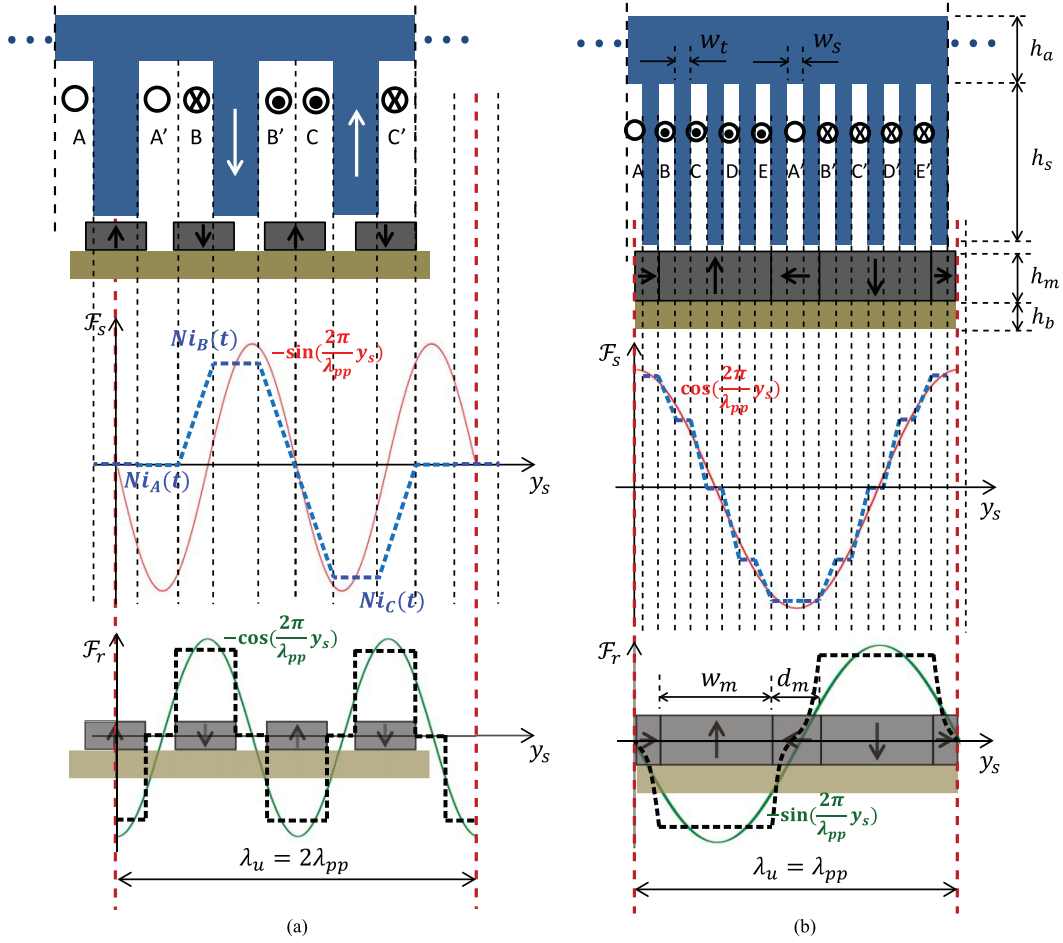


Fig. 1. Magneto-motive force (MMF) waveforms of the stator and moving magnet track (or rotor). The stator MMF, $\mathcal{F}_s$ and rotor MMF, $\mathcal{F}_r$ waveforms are plotted against the stator location, y_s at the given rotor position of $y_r = 0$. Note λ_{pp} is the magnet pole-pair pitch (e.g., from N to N). (a) Conventional linear iron-core motor. (b) Our new fine-tooth motor.

II. LINEAR MOTOR NOISE MECHANISMS

An operating linear iron-core motor experiences two major types of forces between its magnet array and iron-core armature, which are coil-driven force and cogging. Coil-driven force is proportional to the applied coil current. Cogging force, which is not primarily dependent on coil current, is caused by the magnetic interaction between the iron-core teeth (geometric saliency) and the magnets. Depending on the design of a motor, these forces can have significant high-frequency spatial harmonics. Such force harmonics can vibrate the moving stage, which is especially problematic when the stage mechanical eigenmodes are excited. Such stage vibration can be transmitted throughout a precision system structure, and also can radiate as acoustic noise. Both of these can interfere with internal machine elements and with external devices, impairing accuracy.

We have built models for and experimentally measured such linear motor noise sources in a commercially available iron-core linear motor (Tecnotion TL18) and in our new fine-tooth linear motor. The modeling begins with an analysis of armature/magnet force harmonics, as presented in Section III.

III. FORCE HARMONICS OF LINEAR IRON-CORE MOTORS

The motor magnetic force is generated by two interacting magnetic fields, and is proportional to the cross product of the

MMFs of those interacting fields [25] via

$$F_{\text{motor}} \propto \mathcal{F}_s \mathcal{F}_r \sin(\delta_{sr}) \quad (1)$$

where $\mathcal{F}_s$, $\mathcal{F}_r$, and δ_{sr} are the stator MMF amplitude, the rotor MMF amplitude, and the phase difference in the electrical angle between the stator and rotor MMFs, respectively. Note that an electrical angle of 2π corresponds to the magnet pole-pair pitch, λ_{pp} as shown in Fig. 1. The motor force relation in (1) can be rewritten in terms of harmonics (n) as

$$F_{\text{motor}} \propto \sum_{n=1}^{\infty} \mathcal{F}_{s,n} \mathcal{F}_{r,n} \sin(\delta_{sr,n}). \quad (2)$$

Note that in the assumed periodic system, only like-numbered harmonics generate thrust.

A. Conventional 3–4 Combination Motor

The magnetic design of a conventional linear iron-core motor is depicted in Fig. 1(a). The stator armature has three phase windings A, B, and C wound on three iron-core teeth. These three coils excite the armature to magnetically interact with four magnets in the magnet track, as shown in the figure. This type of motor is thus often called a 3–4 combination motor. That is, 3 iron-core teeth and 4 magnets are a fundamental magnetic

unit to generate thrust in such conventional iron-core motors. The length of this basic motor unit is defined as λ_u in the figure. In our moving-magnet design, we refer to the coils and iron as the *stator*, and the magnet array as the *rotor*.

The ideal MMF waves for both the stator and the rotor are sinusoidal as shown with solid curves in Fig. 1(a) so that we have only the fundamentals to generate the thrust without harmonics. More realistic stator and rotor MMF waves are shown in the dashed blue (stator) and dashed black (rotor) curves in the figure. The stator MMF, which can be considered as the magnetic potential ψ_s at the air gap, is estimated with the relation of

$$\mathcal{F}_s = \psi_s = \mathcal{R}\phi = \left(\frac{l}{\mu A}\right) (AB) = \frac{B}{\mu} l = \oint H dl = Ni \quad (3)$$

where $\mathcal{R}$, ϕ , l , A , μ , B , H , and Ni are the magnetic reluctance of an iron-core tooth, the magnetic flux through the tooth, the characteristic length of the flux path, the cross-section area of the flux path, the magnetic permeability of the iron-core material, the magnetic flux density in the flux path, the magnetic field strength in the flux path, and the ampere-turns enclosed by the flux path, respectively.

In the configuration depicted in Fig. 1(a), the first tooth (leftmost) and the first magnet (leftmost) are aligned. We define this rotor position as a reference of $y_r = 0$. To generate thrust on the magnet track in the rightward direction, the three phase currents of the conventional motor are commutated as

$$i_{A,B,C} = I_p \sin\left(\frac{2\pi}{\lambda_{pp}} y_r + \frac{t\pi}{3}\right) \quad (4)$$

where I_p indicates the peak phase current magnitude and t equals 0, 2, and 4 for the phase A, B, and C, respectively. The instantaneous stator MMF values across the teeth can be obtained by multiplying the phase current with the number of winding turns, N as in (3). Note that we use the sign convention where the ampere-turns (Ni) is positive when it creates a magnetic flux downward. The MMF magnetic potential changes linearly across a slot when there are energized coils because the potential change is proportional to the ampere-turns (Ni) in the slot as in (3). The dashed blue curve in Fig. 1(a) depicts the realistic $\mathcal{F}_s$ curve, which can be considered as a sampled version of the ideal sinusoidal waveform (solid red curve) with the iron-core tooth locations as the sampling points. As the realistic $\mathcal{F}_s$ waveform is coarsely sampled, it contains significant harmonics. The realistic rotor MMF $\mathcal{F}_r$ is plotted with the dashed black curve in the figure with the same sign convention. This waveform also contains significant harmonics. Since both the stator and rotor MMFs have such harmonic content, we expect that the magnetic force of the conventional motor also contains high force harmonics via (2).

B. New Fine-Tooth Motor

In order to generate smaller force harmonics, our new motor is designed to have multiple fine teeth with a small tooth pitch so as to *sample* the ideal sinusoidal waveform more densely and thereby create a stator MMF with reduced harmonics. Compared with the conventional iron-core motor design, our new motor has fine teeth, deep slots, and five phases. We therefore refer to our motor as a *fine-tooth* motor. The schematic magnetic design and

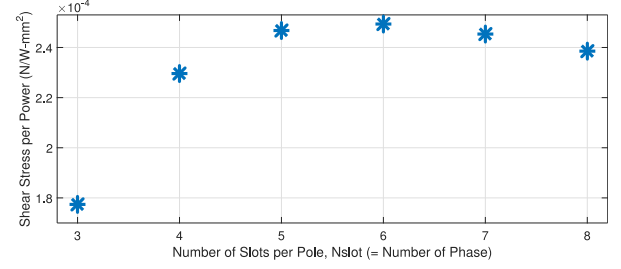


Fig. 2. Shear stress per power versus the number of motor phases. Fine-tooth motor configuration is used and power is calculated over a motor unit, which is one pole-pair.

associated MMF waveforms are illustrated in Fig. 1(b). Note that the realistic stator MMF curve is drawn in the same manner as before, using five-phase sine-commutated phase currents of

$$i_{A,B,C,D,E} = I_p \sin\left(\frac{2\pi}{\lambda_{pp}} y_r - \frac{m\pi}{5}\right) \quad (5)$$

where m equals 0, 1, 2, 3, and 4 for the phase A, B, C, D, and E, respectively.

The magnet array can be either a conventional N-S pattern or a Halbach array, but we choose the Halbach array pattern to produce magnetic fields with smaller harmonics. The rotor MMF is thus also smoother than that of a conventional motor magnet track as can be seen in the figure. Since we have reduced harmonics in both the stator and rotor MMFs, when these interact, we expect smaller force harmonics in the magnetically generated force via (2), and thereby less vibro-acoustic noise with our new motor.

Another advantage of such a fine-tooth motor design is that the cogging force can be significantly reduced relatively easily. Due to the small tooth pitch, the cogging and associated vibro-acoustic noise can be significantly reduced by a small amount of magnet skew. The force harmonic components of our new fine-tooth motor are discussed in Section IV, and the vibro-acoustic noise measurements are presented and compared in Section VI.

IV. NEW FINE-TOOTH MOTOR DESIGN

In this section, we present the design of our new fine-tooth motor and compare its expected motor performance to the conventional 3–4 combination motor. Fig. 1(b) shows key design parameters of a new fine-tooth motor. Since the magnetic features of multiple fine teeth closely configured and Halbach magnet array are expected to provide the force harmonic reduction and hence the motor noise reduction, the design selection process is proceeded mainly to have higher shear stress than the conventional iron-core motor. The detailed design process of our new motor is discussed in [26].

We design our fine-tooth motor to have five phases so as to achieve higher shear stress density per unit power dissipation than a typical 3-phase configuration. Fig. 2 shows a simulated comparison of the shear stress per unit power for different number of phases, with the 5-phase motor enhancing the shear stress per power by 39% compared with the 3-phase motor. Note that for the simulations, the geometric parameters other than the phase number are kept the same. We did not select six or seven phases for our prototype for practical reasons of

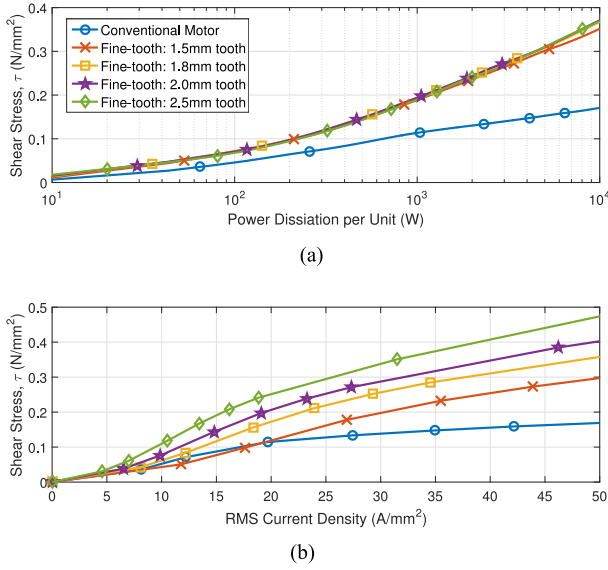


Fig. 3. Simulated performance comparison between the conventional motor and our new fine-tooth motor to select the tooth width value. (a) Shear stress versus power dissipation for various tooth width values. (b) Shear stress versus RMS current density for various tooth width values.

1) a higher attractive force, possibly overpreloading the air bearings in our stage testbed and 2) larger winding overlap, making the winding process more difficult and time-consuming. A 5-phase fine-tooth motor does however require multiple single-phase inverters or a custom-made 5-phase inverter. To avoid such additional effort and to use commercial 3-phase inverters, the 3-phase fine-tooth configuration can also be used to achieve motor acoustic noise reduction compared with the 3–4 combination motors. Alternatively, a 6-phase fine-tooth motor can be used with two 3-phase inverters configured to commutate six phases.

In order to determine geometric design parameters, the shear stress performance is compared over a wide power range and also against the root-mean-square (RMS) current density in the coil wires while changing only the parameter of interest and keeping the other the same, as shown in Fig. 3 for the tooth width, w_t selection. The shear stress, τ in the figure is obtained from the thrust force, calculated by a finite element method, divided by the force-generating area in the working air-gap of a motor. Note that the tooth-to-slot ratio is kept to unity so as to balance the current density in the slot versus magnetic saturation in the tooth, and to minimize cogging.

From Fig. 3(a), we can see our fine-tooth motor having higher shear stress density than the conventional motor at all power levels regardless of the tooth width values. For all design variants, we keep the ampere-turns the same, thereby having the same thrust. However, as the tooth width increases, the force-generating area increases and the coil resistance decreases due to usage of thicker wires for wider slots. This is the reason why we have almost identical curves for all fine-tooth design variants in Fig. 3(a). As can be seen from Fig. 3(b), the fine-tooth motor designs with the tooth width greater than 1.8 mm show higher shear stress than the conventional motor at all current density levels. Among such three cases, $w_t = 2$ mm is selected since it is

TABLE I
FINAL DESIGN PARAMETERS OF OUR NEW FINE-TOOTH MOTOR

Parameter	Description	Value
N_{slot}	Number of slots per pole (# of phase)	5
w_t, w_s	Tooth and slot width	2 mm
h_a	Thickness of armature yoke	15 mm
R_{slot}	Slot aspect ratio (h_s/w_s)	15
N	Number of winding turns per slot	126
AWG	Wire gauge for winding	23
f_{magnet}	Fraction of vertical magnets (w_m/λ_p)	0.7
PM_{thick}	Magnet aspect ratio (h_m/w_m)	0.5
h_b	Magnet back-iron thickness	4.76 mm
D	Motor depth	52 mm
g	Magnetic air gap distance	550 μ m

expected to result in the maximum possible acceleration of our linear stage testbed shown in Section V-B. Note that the other geometric parameters, both for the stator and magnet track, are also determined by the same selection process. The final values of the design parameters are organized in Table I.

Having five phases with the tooth width of 2 mm results in the tooth pitch λ_t and the magnet pole pitch λ_p as

$$\begin{aligned}\lambda_t &= w_t + w_s = 4 \text{ mm} \\ \lambda_p &= N_{slot} (w_t + w_s) = \lambda_u/2 = 20 \text{ mm.}\end{aligned}\quad (6)$$

The slot aspect ratio is selected as $R_{slot} = 15$, and this makes the slot depth as $h_s = w_s \times R_{slot} = 30$ mm. For the magnet track, the sizes of vertical and horizontal magnets are

$$\begin{aligned}w_m &= f_{magnet} \lambda_p = \lambda_p - d_m = 14 \text{ mm} \\ h_m &= w_m \times PM_{thick} = 7 \text{ mm}\end{aligned}\quad (7)$$

where w_m , d_m , and h_m are the width of vertical magnets, width of horizontal magnets, and magnet thickness, respectively, as illustrated in Fig. 1(b). Note that our fine-tooth motor uses relatively larger magnet volume than the conventional motor. However, this is for the purpose of thrust enhancement, and the motor noise reduction comes from the magnetic design of fine teeth and magnet array pattern, regardless of the amount of magnets used.

With the final design parameter values, the expected motor performance is simulated in terms of shear stress and force ripple, and compared with the conventional motor as shown in Fig. 4. We use the same material, M-19 silicon-iron, and so the same magnetic saturation conditions for both motors in the simulation. Note that we use the power dissipation per length for the abscissa in Fig. 4(a). This is to take the motor unit length difference into account to have a fair comparison between two different motors. As can be seen from the shear stress comparison, our new fine-tooth motor (solid red curve) shows higher shear stress than the conventional motor (dashed blue curve) at every power level. Specifically at a lower power level of 10 W/mm, our fine-tooth motor is expected to generate a 28% higher shear stress of 0.115 N/mm² than the conventional motor's shear stress of 0.090 N/mm². At a higher power level

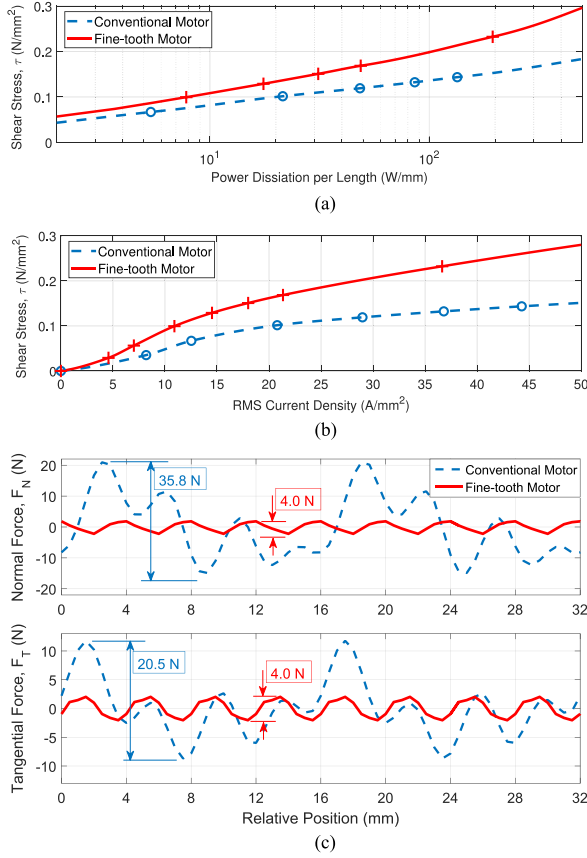


Fig. 4. Simulated performance comparison between the conventional motor and our new fine-tooth motor with final design parameters. (a) Shear stress versus power dissipation per motor unit. (b) Shear stress versus RMS current density in the coil wires. (c) Force ripple in normal and tangential directions.

of 500 W/mm where the material saturation is expected, we observe even higher shear stress increase of 62% from 0.183 to 0.296 N/mm², showing significant force potential of our new fine-tooth motor. This shows that the saturation is relatively less problematic in the fine-tooth motor design because, although narrow, multiple teeth compactly arranged help distribute the magnetic flux more uniformly.

We also compare the shear stress performance between the two motors against the RMS current density in the coil wires, as shown in Fig. 4(b). This is an important comparison because the RMS current density in the wire is directly related to the thermal limit of a motor. As can be seen from the figure, our fine-tooth motor with the final design parameters also shows higher shear stress than the conventional motor for the same thermal condition.

Fig. 4(c) compares the force ripple of the fine-tooth motor with the conventional motor. As shown in the figure, our new motor generates smaller force ripple. Specifically, we have a significant peak-to-peak force ripple reduction with the ratio of 9-to-1 (from 35.8 to 4.0 N) and 5-to-1 (from 20.5 to 4.0 N) in the normal (Z) and tangential (Y) directions, respectively. We can also observe that our new motor contains less force harmonics than the conventional motor. This is consistent with the discussion in Section III. The conventional motor has a strong fundamental harmonic (with a period of 16 mm, same as

the tooth pitch) along with other high harmonics of significant amplitudes. On the other hand, our new fine-tooth motor exhibits almost solely the fundamental harmonic (with a period of 4 mm) without any notable high harmonics due to the smoother stator and rotor MMF's as shown in Fig. 1(b). We expect that the advantage of having less force harmonics leads to less motor noise as discussed in Section II. The relevant experimental results to validate this expectation and to show how much less noise we can have with our new fine-tooth motor are detailed in Section VI.

V. EXPERIMENTAL TESTBED

We introduce in this section an experimental linear stage testbed designed and constructed for the motor vibro-acoustic noise experiments. In this research work, we use the magnetic configuration of moving short magnet with stationary long armature because 1) it facilitates a stage acceleration increase due to the moving mass reduction so that we can investigate the motor noise issue at a high acceleration, and 2) the moving magnet minimizes the number of umbilical cables that might cause assembly complexity and also undesired cable dynamics.

A. Two Different Iron-Core Motors

Fig. 5(a) shows a conventional 3-phase linear iron-core stator, TL18 by Tecnotion, with a total of 18 lumped phase windings. The moving magnet track has a conventional array pattern of N-S-N-S as shown in Fig. 5(c). We use both nonskewed and skewed magnet tracks to investigate the effect of skewing on the vibro-acoustic noise. Fig. 5(b) shows our new fine-tooth motor stator. Its length is designed to be longer than the conventional motor to have a larger travel range. The armature consists of a total of 180 coils where each prewound coil is inserted in slots with the double-layered pattern and winding coils of each phase are wired in series. This is a relatively difficult process compared with the lumped windings in the conventional motor design. Manufacturability improvement of the fine-tooth motor will be pursued as a future work, and will not be addressed herein. Note that the stator armature is divided in four sections to distribute the power, so we use a total of 20 power amplifiers (5 phases/section $\times$ 4 sections) to drive our motor so as to fit within power limits of readily available amplifiers. The magnet tracks of our new motor are shown in Fig. 5(d). Both nonskewed and skewed magnets are assembled in the Halbach pattern with the vertical magnet portion of 70% out of a pole pitch, as discussed in Section IV. The skew angle is determined by

$$\theta_{\text{skew}} = \text{atan} \left(\frac{\lambda_t}{D} \right) = \text{atan} \left(\frac{4 \text{ mm}}{52 \text{ mm}} \right) \simeq 4.4^\circ \quad (8)$$

in order to cover one full tooth pitch λ_t with a given motor depth D .

B. Linear Stage Testbed

Our experimental linear stage testbed is shown in Fig. 6. Note that for the noise tests of the conventional motor, the same setup is utilized with only the stator and magnet track changed. We use air bearings for the moving stage to avoid the noise generation

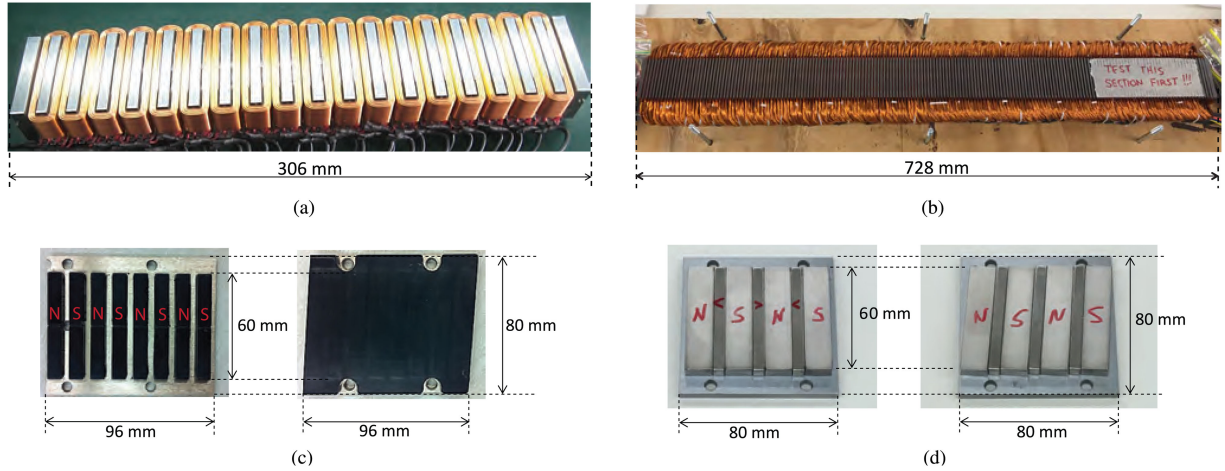


Fig. 5. Pictures of linear iron-core permanent magnet motors. (a) Conventional stator armature. Photo courtesy of Tecnotion. (b) Fine-tooth stator armature. (c) Nonskewed (left) and skewed (right) magnet tracks of conventional motor. Skew angle of 4° . (d) Nonskewed (left) and skewed (right) magnet tracks of our fine-tooth motor. Skew angle of 4.4° .

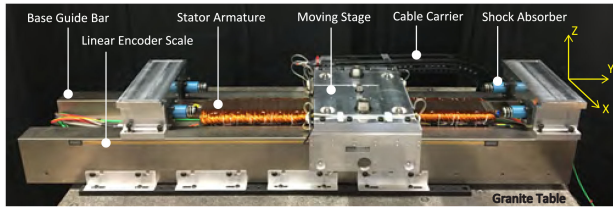


Fig. 6. Completed linear stage testbed with our new fine-tooth motor. Magnet track is mounted on the bottom surface of the moving stage.

by mechanical contact. In order to suppress undesired cable dynamics and reduce associated vibrations and noises, a link-free and hinge-free cable carrier (Quantum 040-28 by Tsubaki) is used. Shock absorbers (SCS33-25 by ACE) are installed at both ends of travel to be able to safely and passively stop the stage in an emergency runaway situation. Optical linear position sensors (read-head T1001 and scale RGSZ20 by Renishaw) are used to provide a real-time position feedback to the position controller. The interpolated resolutions can be chosen to be either $1 \mu\text{m}$ or $0.1 \mu\text{m}$. A dynamometer configured by four triaxial load cells (Type 9250A4 by Kistler) under the stator directly measures the motor forces (e.g., cogging). We use accelerometers (353B15 and 356A26 by PCB Piezotronics) and microphones (130E20 by PCB Piezotronics) to measure both vibrational and acoustic noises emitted by the motors.

The stage is position-controlled at the -3 dB-bandwidth of about 200 Hz. The position feedback control loop runs deterministically at 10 kHz in a real-time controller (PXI-8110 by National Instruments). Another real-time processor (PXIe-8133 by National Instruments) controls a measurement loop to measure both vibrational and acoustic noises at the sample rate of 50 kHz.

VI. EXPERIMENTAL RESULTS

In this section, we present experimental results on the vibro-acoustic noise of the conventional linear iron-core motor and our new fine-tooth motor. We begin to discuss the motor noise

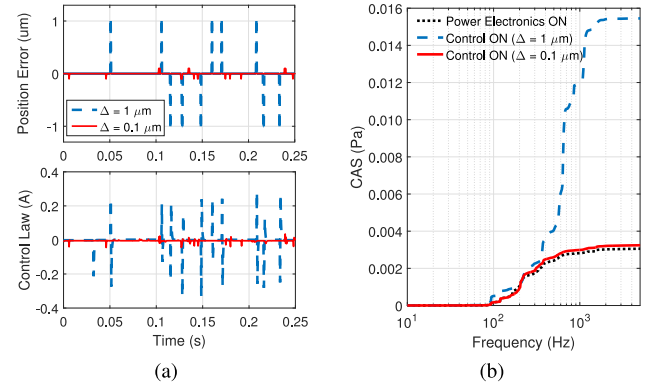


Fig. 7. Motor noise in stationary position. (a) Encoder quantization error and corresponding control effort with $\Delta = 1 \mu\text{m}$ (dashed blue) and $0.1 \mu\text{m}$ (solid red). (b) Comparison of acoustic noise CAS.

issue in a stationary status, and then present the motor noise in cyclic operations. The experimental data are compared to show how our new fine-tooth motor outperforms the conventional 3–4 combination motor in terms of vibro-acoustic noise.

A. Noise in Stationary Position

Before operating our stage in a cyclic mode, we observe serious vibro-acoustic noise when the stage is commanded to a stationary position. We find it stems from the encoder quantization together with a high loop gain, exerting impulse-like forces on the stage. Fig. 7(a) shows the position error, when commanded to hold a stationary position, by the encoder quantization, Δ of $1 \mu\text{m}$ (dashed blue line), and resultant control effort generated by our high-gain position controller. As shown in the figure, control effort peaks are substantial because of significant quantization errors. Such impulse-like control efforts generate vibration and acoustic noise as can be seen from the cumulative amplitude spectra (CAS) of noise measurement in Fig. 7(b). Note that our noise baseline is when PWM drives are turned ON as shown with dotted black curve in the noise CAS plot. With $\Delta = 1 \mu\text{m}$, we observe significantly increased noise

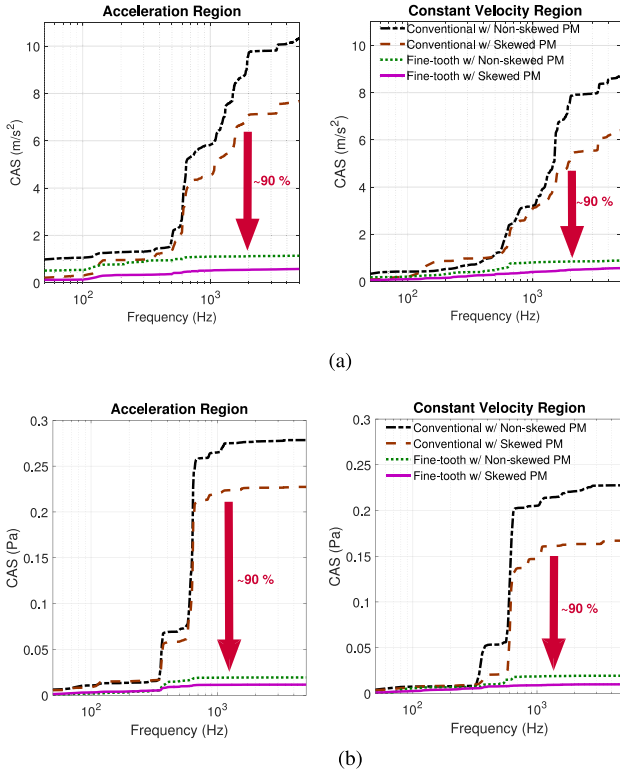


Fig. 8. CAS comparison of measured noise between conventional motor and our new fine-tooth motor, showing significant noise reduction during both acceleration (left) and constant velocity (right) regions. (a) Vibrational noise. (b) Acoustic noise.

from the baseline, which comes from the stage eigenmodes, specifically a bending mode of our U-shaped stage at 630 Hz.

In order to resolve this issue, we changed hardware to increase the interpolation ratio to have a ten times finer encoder resolution of $0.1 \mu\text{m}$. Fig. 7(a) shows with solid red curves the quantization error and its associated control effort with $\Delta = 0.1 \mu\text{m}$. Due to much smaller position errors, the control effort peak is substantially reduced. This leads to significant reduction of motor stationary acoustic noise as shown with the solid red curve in Fig. 7(b). The overall stationary noise CAS is decreased closely to the noise baseline. Specifically, we achieve 79% reduction of the acoustic noise over the bandwidth of 5 kHz.

For high-accuracy servo applications, required position or velocity error tolerances can be fulfilled even with a relatively coarse encoder resolution, such as $\Delta = 1 \mu\text{m}$ in our case. However, the encoder quantization and its effects have to be also taken into account when choosing an encoder resolution for a high-precision system sensitive to noise.

B. Noise During Cyclic Operation

Here, we discuss the motor vibro-acoustic noise during cyclic operations. Note that the encoder resolution of $0.1 \mu\text{m}$ is used throughout this subsection and hereafter in order to minimize the quantization issue. Fig. 8(a) and (b) show the CAS comparison of vibrational and acoustic noises between conventional motor and our new fine-tooth motor when either nonskewed or skewed magnets are used. The motor noise is measured while the stage is cycled following a fourth-order position trajectory

with a maximum acceleration A_{max} of 25 m/s^2 and a maximum velocity V_{max} of 1 m/s .

When the stage is driven by the conventional motor, we observe several eigen-frequencies excited, such as at 360 Hz, 500 Hz, 630 Hz, 1100 Hz, etc., dominating the overall noise level in both accelerometer and microphone data during both acceleration and constant velocity periods. These stage dynamics are excited by the ample force harmonics of the conventional linear motor because, by design, its stator and rotor MMF's contain high spatial harmonics as discussed in Section III. Note that the measured vibro-acoustic noise level is a bit smaller with the skewed magnets. This is because skewing magnets acts as a low-pass filter to smoothen the rotor MMF, thereby generating less force harmonics.

Compared with the conventional 3–4 combination motor, our fine-tooth motor produces less force harmonics via its fine-tooth magnetic design as compared in Fig. 1. Therefore, we observe in Fig. 8 much less overall noise during not only acceleration regions, but also constant velocity periods. Table II compares the overall noise level, accumulated within the bandwidth of 5 kHz, during both acceleration and constant velocity regions for each of all four cases of motor types. Note that the cumulative amplitude of vibration noise has units of m/s^2 and the acoustic noise is in pascals and decibels. The decibel values in parentheses indicate the sound pressure level (SPL) calculated as

$$\text{SPL} = 10 \log_{10} \left(\frac{p}{p_{\text{ref}}} \right)^2 \quad (9)$$

where p and p_{ref} are the sound pressure measured by microphones and standard reference sound pressure of $20 \mu\text{Pa}$, respectively. With our new fine-tooth motor, the overall vibrational noise during acceleration with nonskewed magnets is reduced from 10.31 to 1.14 m/s^2 , showing 89% noise reduction. The acoustic noise during acceleration also shows a significant noise reduction from 0.278 Pa ($\approx 83 \text{ dB}$) to 0.019 Pa ($\approx 60 \text{ dB}$), showing 93% reduction in Pascals and 23 dB reduction in SPL. Qualitatively speaking, this is the noise difference between a truck passing by and a quiet office, i.e., a large reduction. These significant overall noise reductions are also observed during constant velocity regions for both vibrational and acoustic noises, specifically 90% and 92% with nonskewed magnets, respectively. This is because our fine-tooth motor contains, by design, less force harmonics so that the stage dynamics are not significantly excited in either the acceleration or constant velocity region. Specifically, the major stage eigenmodes at 360 Hz, 500 Hz, and 630 Hz are less excited as can be seen by the smaller jumps at those frequencies from the CAS's in Fig. 8(a) and (b).

C. Noise Associated With Cogging

Cogging-related noise is dominant during constant-velocity regions where the motor coil currents are low. Note that, although small, nonzero phase currents are required to overcome the cogging, thereby producing nonzero coil-driven force ripple in constant-velocity regions. However, these force ripples are a second-order term, and we use the term “cogging” in this section because it is the cogging that is a major portion of force fluctuation and the culprit for vibro-acoustic noise in constant-velocity

TABLE II
OVERALL VIBRO-ACOUSTIC NOISE LEVEL COMPARISON BETWEEN THE CONVENTIONAL MOTOR AND OUR NEW FINE-TOOTH MOTOR

	Vibrational Noise (m/s ²)		Acoustic Noise (Pa)	
	Acceleration Region	Constant Velocity Region	Acceleration Region	Constant Velocity Region
Conventional (Non-skew)	10.31	8.71	0.278 ($\approx$ 83 dB)	0.228 ($\approx$ 81 dB)
Conventional (Skew)	7.67	6.41	0.227 ($\approx$ 81 dB)	0.167 ($\approx$ 78 dB)
Fine-tooth (Non-skew)	1.14	0.89	0.019 ($\approx$ 60 dB)	0.019 ($\approx$ 60 dB)
Fine-tooth (Skew)	0.58	0.56	0.012 ($\approx$ 55 dB)	0.010 ($\approx$ 54 dB)

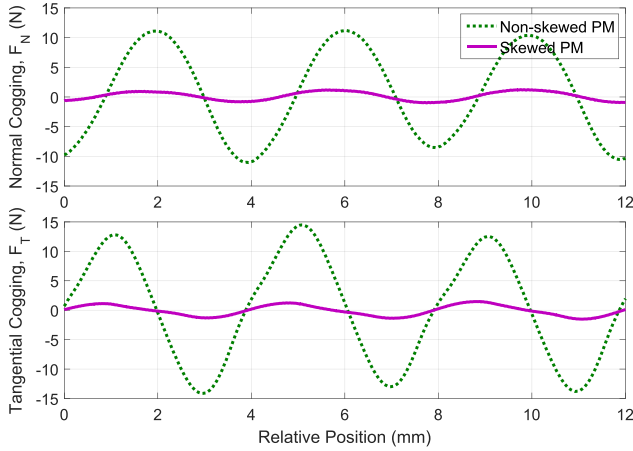


Fig. 9. Fine-tooth motor cogging force measurement comparison between nonskewed and skewed magnets. Cogging fundamental period equals a tooth pitch, $\lambda_t = 4$ mm.

regions. In our fine-tooth design with a small tooth pitch, the cogging can be significantly reduced by skewing the magnets. Fig. 9 shows the measured cogging forces of our fine-tooth motor with either the nonskewed or skewed magnets. As shown in the figure, skewing significantly reduces the cogging in both the normal (Z) and tangential (Y) directions. Specifically, the peak-to-peak amplitudes are reduced from 22.26 N (normal) and 28.62 N (tangential) to 2.22 N and 2.99 N, respectively. This is about a 10-to-1 (90%) reduction in both directions.

This significant cogging reduction directly leads to the motor noise reduction. In order to investigate the cogging-driven noise, we command the stage to follow a trajectory with an increasing constant velocity level at each cycle, namely from 1.6 to 2.2 m/s by 0.2 m/s while measuring the vibro-acoustic noise along the cycles. The expected frequency of the cogging-oriented noise can be written as

$$f_{\text{cogging}} = \frac{v_c}{\lambda_{\text{cogging}}/n} \text{ (Hz)} \quad (10)$$

where v_c , λ_{cogging} , and n are the constant velocity, fundamental cogging period, and cogging harmonic index, respectively. Note that the fundamental cogging period is the same as the tooth pitch, $\lambda_{\text{cogging}} = \lambda_t = 4$ mm, and the harmonic index n is equal to one since the cogging of our new fine-tooth motor is mainly the fundamental as shown in Fig. 9.

Using the noise measurement, we obtain spectrograms to observe the spectral components of the noise over time. Fig. 10(a) shows a spectrogram of acoustic noise with nonskewed

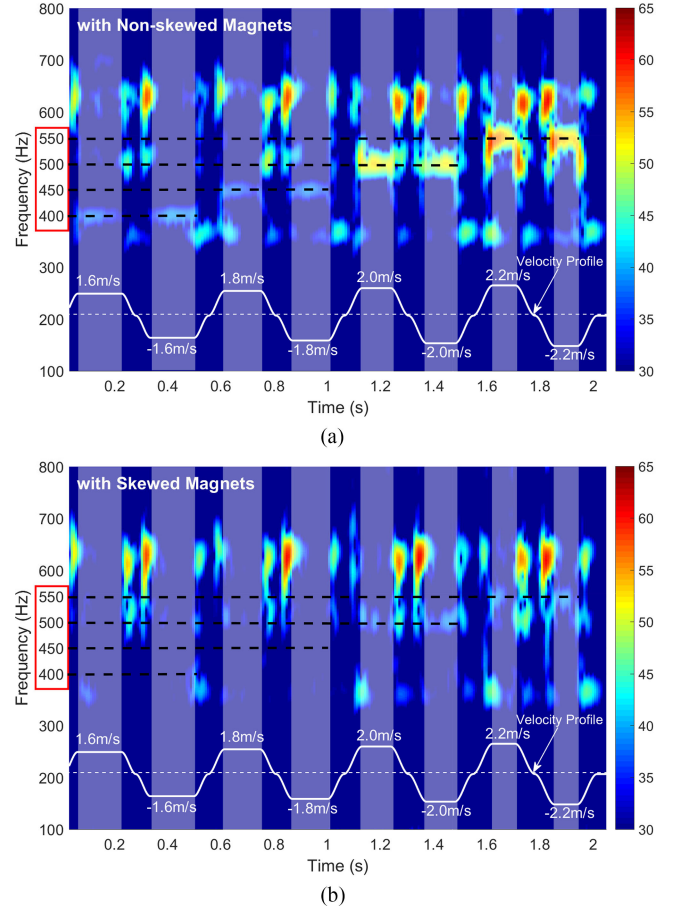


Fig. 10. Spectrogram of acoustic noise of new fine-tooth motor during increasing-velocity cycles with $A_{\text{max}} = 45$ m/s² and $V_{\text{max}} = 1.6$ –2.2 m/s. The color maps show the SPL in decibels. (a) Noise with nonskewed magnets. (b) Noise with skewed magnets.

magnets. Note that the velocity profile commanded for the stage to follow is overlapped in the plot. The bright regions in the figure illustrate the constant velocity regions while the dark regions are for the acceleration and deceleration periods. In the spectrogram, we observe the cogging-oriented noise dominating the constant velocity regions at increasing frequencies of 400 Hz, 450 Hz, 500 Hz, and 550 Hz for 1.6 m/s, 1.8 m/s, 2.0 m/s, and 2.2 m/s, respectively, as predicted in (10). When we hear this noise in person, it sounds as if the stage plays a musical scale with increasing pitch. Note that we observe that the cogging noise intensity increases when the cogging noise frequency becomes closer to the eigen-frequency of 630 Hz. An interesting observation found during acceleration and deceleration regions

is the zero-noise points. At every half cycle, the velocity flips its sign as shown from the velocity profile in Fig. 10(a). At those points of sign-flipping, the velocity and acceleration instantaneously become zero, associated with a momentary stop of the moving stage at the turnaround point. We observe no noise at those points, which supports our hypothesis that the motor acoustic noise comes from the moving part vibrated by magnetic force fluctuations.

This cogging-driven noise is significantly reduced by using skewed magnets, as shown in Fig. 10(b). While there is still motor noise associated with stage dynamics during acceleration and deceleration regions, we see that the cogging noise during constant velocity regions is significantly reduced, compared with the case with nonskewed magnets. For 1.6 m/s and 1.8 m/s, the cogging noises at 400 Hz and 450 Hz are almost completely eliminated such that it is not noticeable within the chosen SPL range. For the higher velocities of 2.0 m/s and 2.2 m/s, we observe a bit of cogging noise at 500 Hz and 550 Hz, but these are at a smaller level than the case with the nonskewed magnets. Specifically, the cogging noise at 550 Hz is reduced from 60 to 40 dB by skewing. This SPL decrease of 20 dB corresponds to a 10-to-1 (90%) noise reduction in Pascals. This significant noise reduction is achieved simply by skewing the magnets (only 4.4°) due to our new fine-tooth motor design with a short tooth pitch.

VII. CONCLUSION

This paper described the design and testing of a new fine-tooth linear iron-core permanent magnet motor to achieve low vibro-acoustic noise while also enhancing thrust capability. We presented in this paper studies of the vibro-acoustic noise issue of linear iron-core synchronous motors. It is hypothesized and experimentally proven that such motor noise is caused by high force harmonics vibrating the moving stage, and is especially severe when the stage eigenmodes are excited. Such stage vibration can be transmitted through the system structure and can radiate as acoustic noise, thereby disturbing precision machines.

While significantly reducing the motor noise, our new fine-tooth motor also outperforms the conventional 3–4 combination motor in the aspect of shear stress density. At a power-per-motor-unit-length level of 10 W/mm, our new motor is expected to produce a 28% higher shear stress than the conventional iron-core motor, and a larger increase of 62% at higher power level of 500 W/mm.

Looking to next steps, we expect that the motor noise can be further reduced by mapping the residual force fluctuation in the normal- and shear-directions, and by modifying the commutation laws to cancel this force in the open-loop sense. We also envision using a double-sided motor configuration where an additional stator and magnet track are used in a mirrored fashion. By symmetrically configuring the identical motors in the lower and upper positions, we can cancel out the normal forces and thus expect further noise reduction. The double-sided motors are now being studied and will be presented in detail in a future publication.

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